Project Documentation

Project Title: LibSpace-A Robust Backend for Library Operations

1. **Introduction**:
   - **Project Title:**LibSpace-A Robust Backend for Library Operations
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2. **Project Overview:**

Purpose:

The purpose of Libspace is to develop a backend system that manages library book records and user accounts securely.It demonstrates user

authentication,secure API development,database integration,and CRUD operations.

Features:

- User registration and login
- Secure password hashing using bcrypt
- JWT-based authentication
- Add,update,and delete books
- View book list and get book by ID
- One user can manage multiple books

3. Architecture:

API Layer(Routes)

Handles incoming HTTP requests and API endpoints using Node.js and Express.js.

- /api/auth/register
- /api/auth/login
- /api/book/add
- /api/book/:id

Business Logic Layer(Controllers)

- Controllers implement core application logic including authentication,password hashing,JWT generation,and CRUD operations.

Data Access Layer(Database/Models)

- The data layer manages database interaction using Mongodb and Mongoose.

- Models represent Users and Books collections.

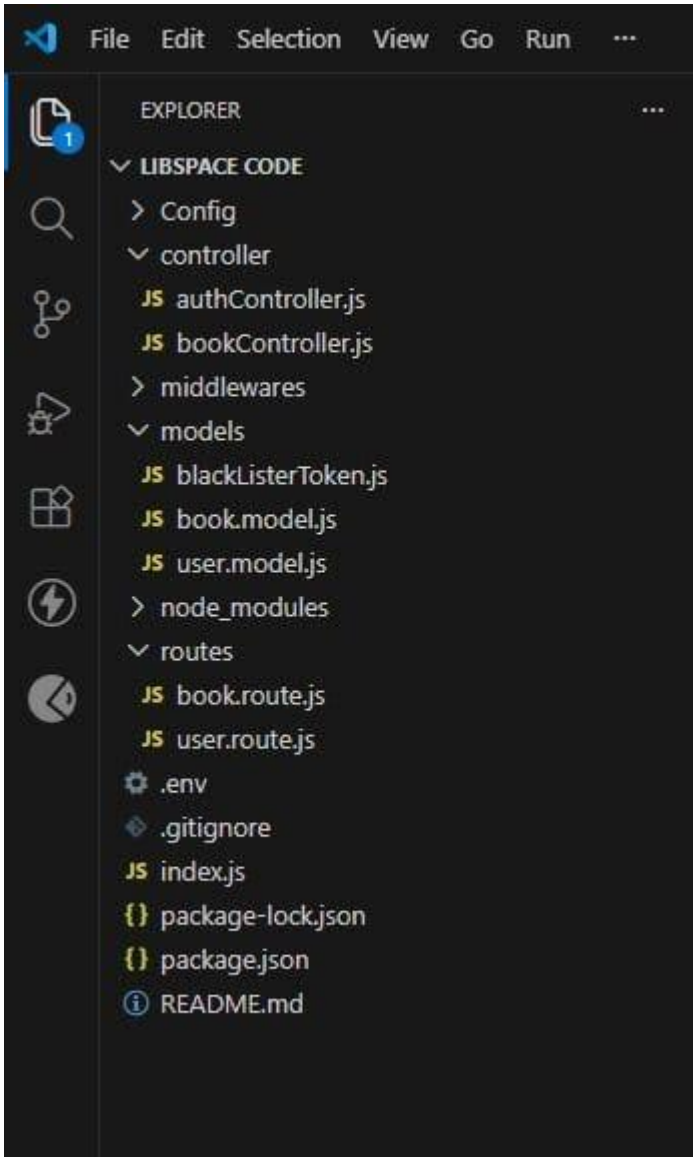
4. Technologies Used:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT(JSON Web Token)
- bcrypt
- Thunder client
- Visual Studio Code

5. Installation:

- Install Dependencies
  npm install
- Environment Variables(.env)
  PORT=5000
  MONGO_URI=mongodb;//127.0.0.1;27017/libs pace
  JWT_SECRET=your_secret_key
- Run the Server
  npm start
  or
  npm run dev

6. Folder Structure:



7. **API Documentation:**

Users APIs

- POST/api/auth/register
- POST/api/auth/login
- DELETE/api/auth/delete
- POST/api/auth/logout

Book APIs

- POST/api/book/add
- GET/api/book
- GET/api/book/:id

- PUT/api/book/:id
- DELETE/api/book/:id

8. **Authentication:**

LibSpace uses JWT for authentication.After login,the server generates a token which must be sent in request headers for protected routes.

Authorization: Bearer <token>

9. **Database Design**

   Users Collection Fields

   - name
   - email
   - password
   - createdAt
   - updatedAt

   Books Collection Fields

   - title
   - author
   - url
   - user(Objected reference)

10. **Testing:**

    1. Tested using Thunder client
    2. Verified using registration and login
    3. Verified book CRUD operations
    4. Verified token authentication

11. **Screenshots:**

Fig 1.1 Successful Connection to MongoDB

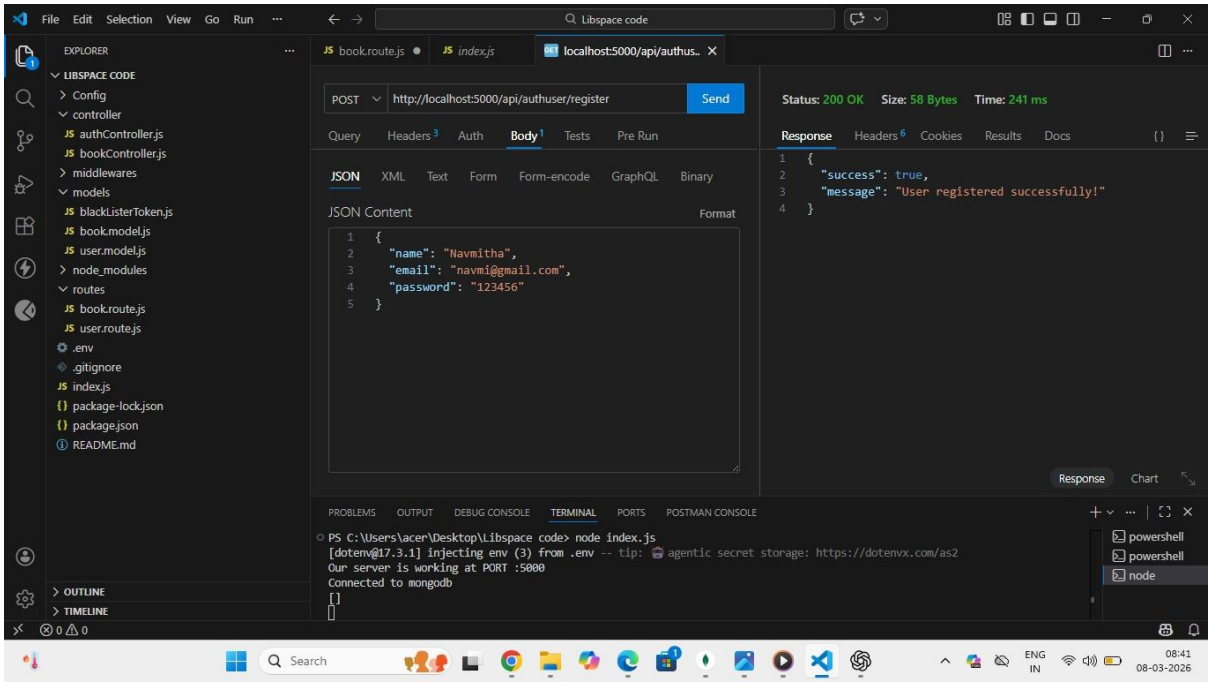


Fig 1.2 Displaying Book Title and Author

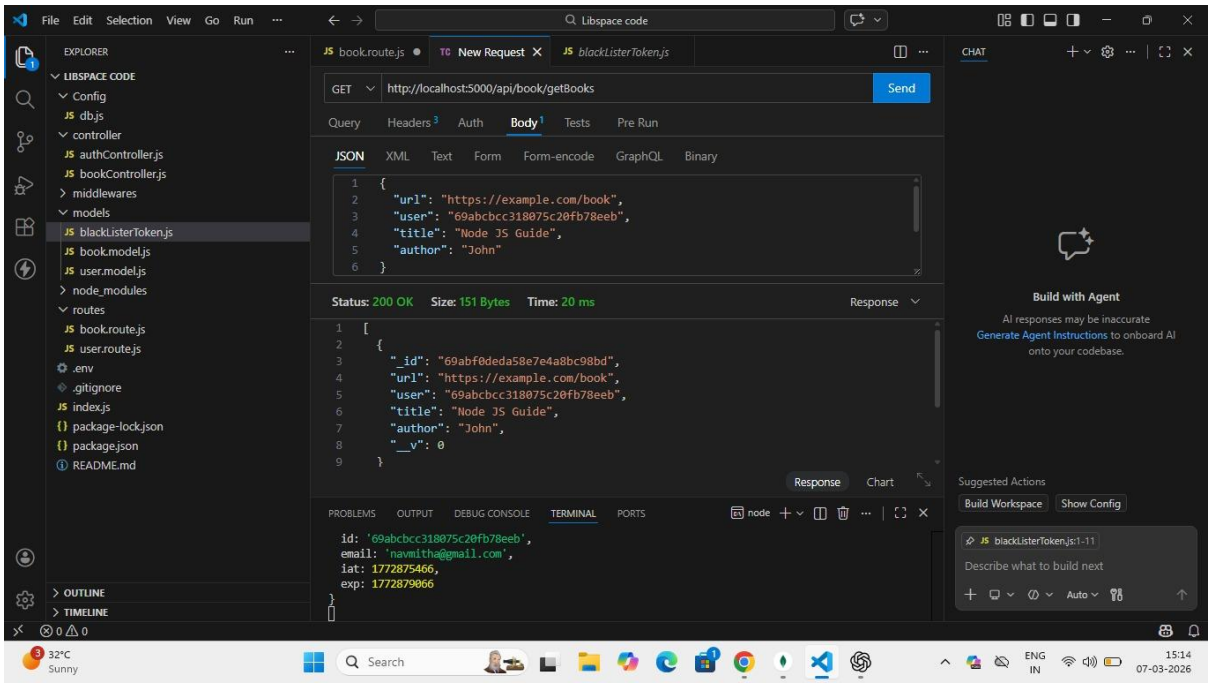


Fig 1.3 Adding a New Book

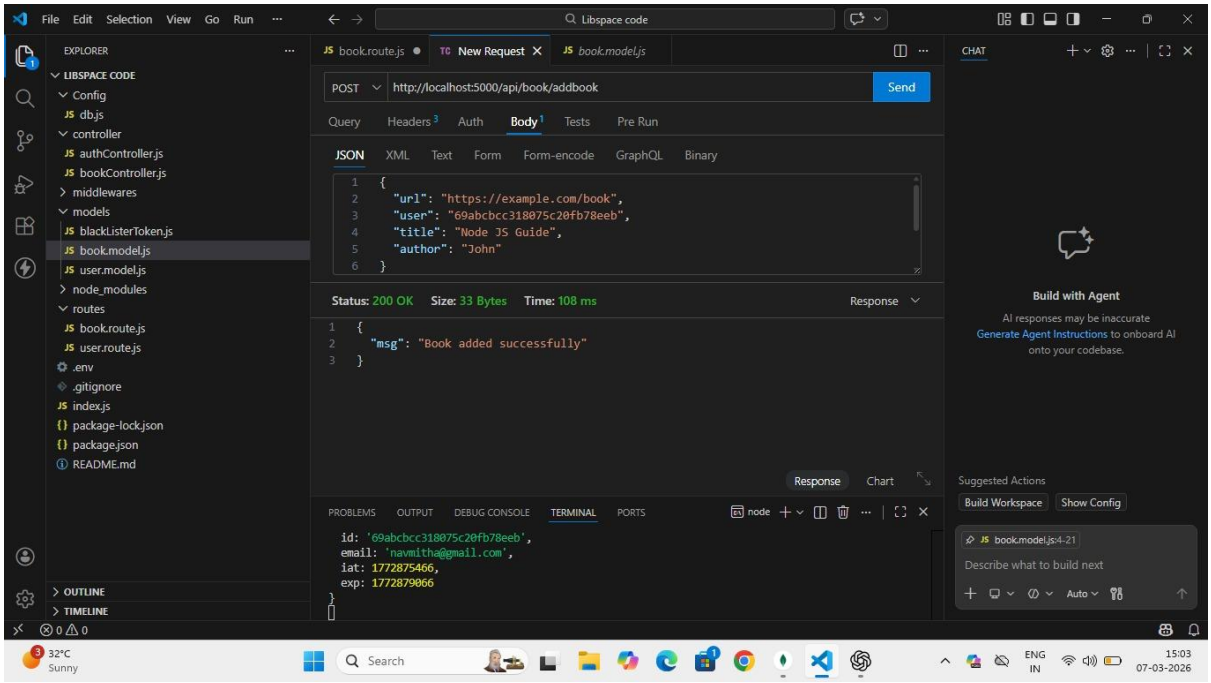


Fig 1.4 Deleting a Book

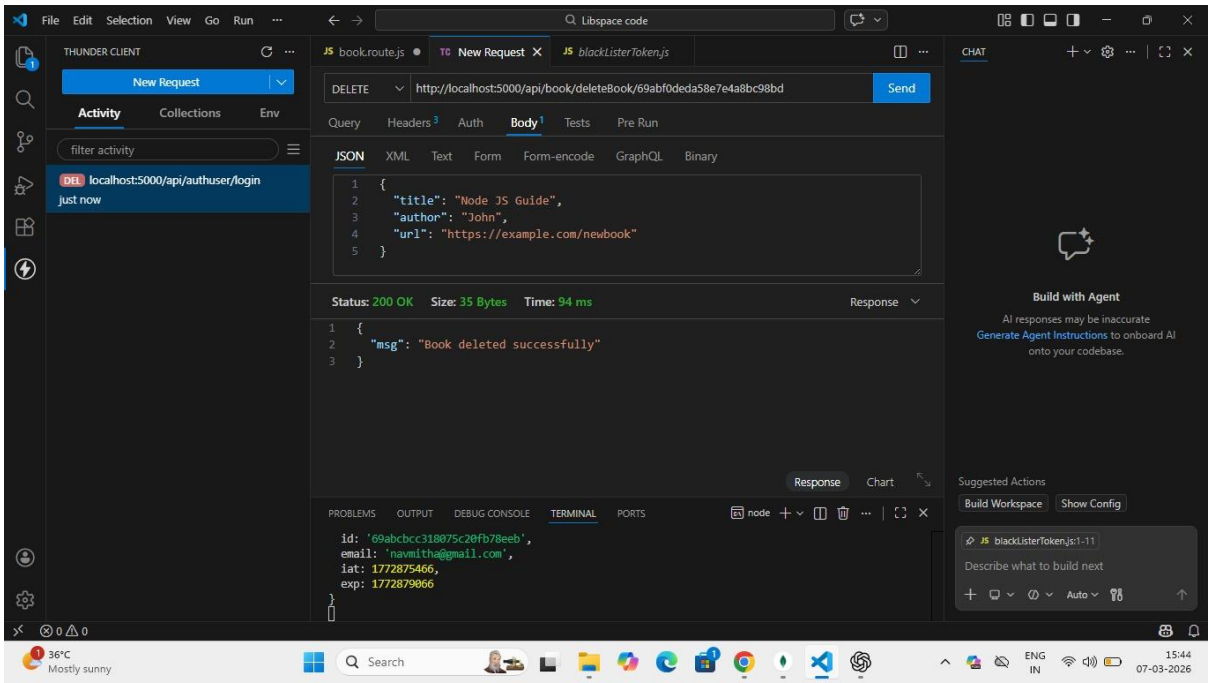
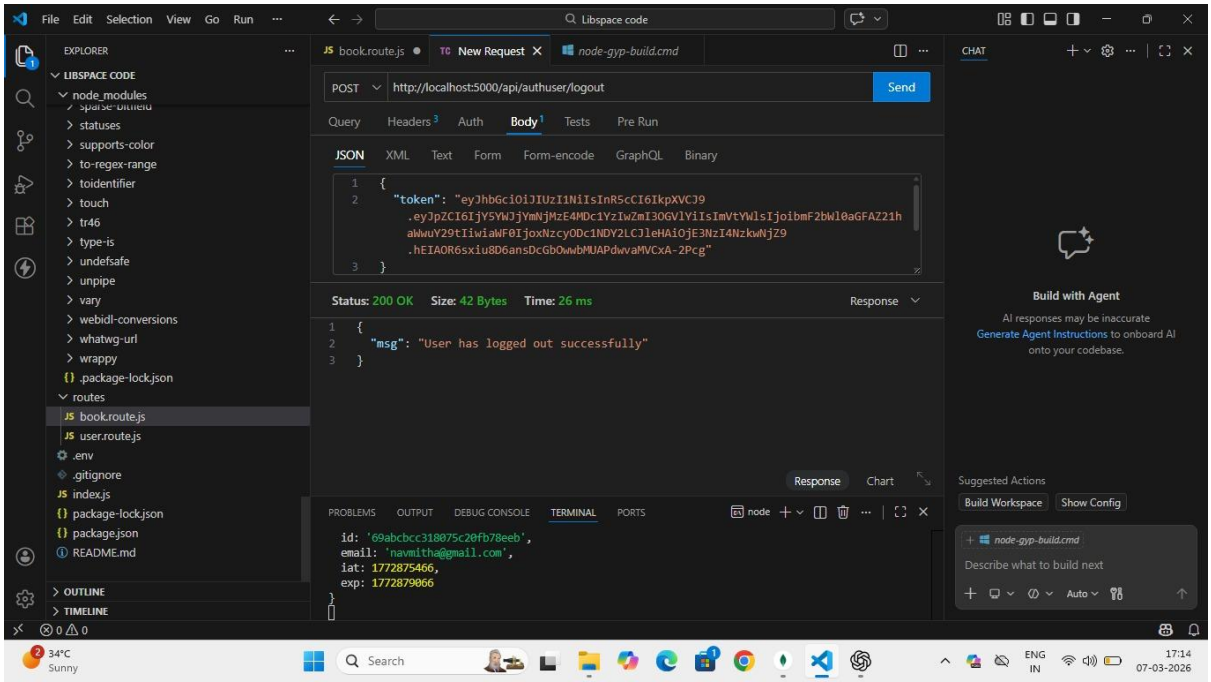


Fig 1.5 User Logout Response



12.Known Issues:

- No frontend interface
- Requires Thunder client for testing
- Token blacklist optimization needed

13.Future Enhancements:

- Frontend integration with React
- Role-based authentication
- Cloud database deployment
- Book file upload support
- Mobile application support

14.Demo Video Link:

https://youtu.be/dJe0bknNF9Y?si=6aBmQu5uzZIHWTjQ

15.Libspace Project Code:

https://drive.google.com/drive/folders/1_bYY6CaN1uWB6pdFYpwfyJqSS0CgI-MO